



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2006
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK # 2

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes.
- Working time – 2 hours.
- Write using black or blue pen.
- Board approved calculators may be used.
- All *necessary* working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for messy or badly arranged work.
- Hand in your answer booklets in 3 sections.
Section A (Questions 1 - 3),
Section B (Questions 4 - 6) and
Section C (Questions 7 - 8).
- Start each **NEW** section in a separate answer booklet.

Total Marks - 90 Marks

- Attempt Sections A - C
- All questions are NOT of equal value.

Examiner: *E. Choy*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

Total marks – 90
Attempt Questions 1 - 8
All questions are NOT of equal value

Answer each section in a SEPARATE writing booklet. Extra writing booklets are available.

SECTION A (Use a SEPARATE writing booklet)

Question 1 (8 marks)	Marks
(a) Given $z = \frac{3-i}{2+i}$ find	
(i) $z\bar{z}$	2
(ii) $\tan(\arg z)$	1
(b) (i) If $z = 1 + \cos\theta + i\sin\theta$, for $0 < \theta < \pi$, then show that	3
$z = 2 \cos \frac{\theta}{2} \left(\cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \right)$	
(ii) Hence, find $ z $ and $\arg z$ in terms of θ	2
Question 2 (9 marks)	
(a) If α, β and γ are the roots of $x^3 - 2x + 5 = 0$, find the cubic polynomial that has roots	
(i) $2 - \alpha, 2 - \beta$ and $2 - \gamma$.	2
(ii) $\alpha^2 + \beta^2, \beta^2 + \gamma^2$ and $\gamma^2 + \alpha^2$	2
(b) Given that p and q are real and also that $1 - 4i$ is a root of the equation	
$x^2 + (p+i)x + (q-5i) = 0$	
(i) Find the values of p and q .	3
(ii) Find the other root of the equation.	2

Question 3 (13 marks)

Marks

(a) Let $\omega = \cos \theta + i \sin \theta$, where $0 < \theta < \pi$

(i) Express ω^2 and $\frac{1}{\omega}$ in modulus-argument form. 2

(ii) Given that $\omega^2 + \frac{5}{\omega} - 2$ is a purely imaginary number, show that 3
 $2 \cos^2 \theta + 5 \cos \theta - 3 = 0$.

(iii) Hence, or otherwise, evaluate θ and express ω in modulus-argument form. 2

(b) A and B are two distinct points in an Argand diagram representing two distinct, non-zero numbers z_1 and z_2 respectively.
 Suppose that $z_2 = \omega z_1$, where ω is the number found in (a) above.

(i) Find $\left| \frac{z_2}{z_1} \right|$ and $\arg \left(\frac{z_2}{z_1} \right)$ 3

(ii) Show that $\triangle AOB$ is an equilateral triangle, where O represents the number 0 in the Argand diagram. 3

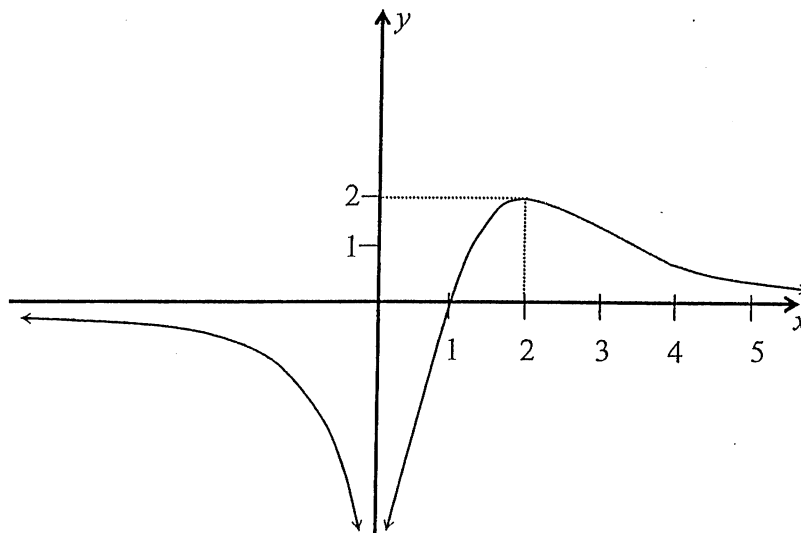
End of Section A

SECTION B (Use a SEPARATE writing booklet)

Question 4 (12 marks)

Marks

The sketch of $y = f(x)$ is shown below



You are given that the x – intercept is $x = 1$;
 $x = 0$ is a vertical asymptote and $y = 0$ is a horizontal asymptote.
 As well, $(2, 2)$ is a stationary point and $(4, 1)$ is a point of inflexion.

On the Answer Sheet provided draw separate graphs for

- | | | |
|-------|--|---|
| (i) | $y^2 = f(x)$ | 2 |
| (ii) | $y = \frac{1}{[f(x)]^2}$ | 3 |
| (iii) | $y = f'(x)$ | 2 |
| (iv) | $y = e^{f(x)}$ | 2 |
| (v) | Draw a possible diagram for the primitive function of $f(x)$ | 3 |

Question 5 (10 marks)

Marks

- (a) Given that C is the curve defined by

$$f(x) = \frac{\sin 2x}{2 - \cos 2x}$$

where $0 \leq x \leq \pi$

- (i) Find the x and y intercepts of C .
 (ii) Find the turning point(s) of C and determine their nature.
 (iii) Show that $f(x) = -f(\pi - x)$.

2

3

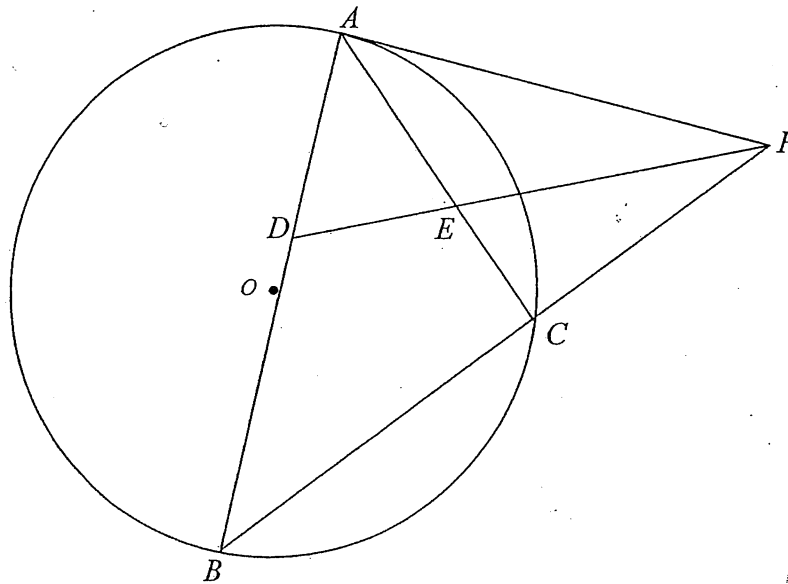
2

- (iv) Sketch the curve $y = \frac{|\sin 2x|}{2 - \cos 2x}$ for $0 \leq x \leq \pi$

3

Question 6 (8 marks)

- (a) In the diagram below, PA is the tangent to the circle at A , whose centre is O .
 Line PCB cuts the circle at B and C .
 The angle bisector of $\angle APB$ meets AB at D and AC at E .



Prove that $\frac{DB}{AB} + \frac{EC}{AC} = 1$

6

- (b) How many ways are there to pick a man and a woman who are not husband and wife from a group of n married couples?

2

End of Section B

SECTION C (Use a SEPARATE writing booklet)

Question 7 (21 marks)

Marks

(a) (i) Find $\int \frac{e^x}{\sqrt{1-e^{2x}}} dx$ 2

(ii) Find $\int \frac{3x^2 - 6x + 1}{(x-3)(x^2+1)} dx$ 2

(iii) Find $\int \frac{1}{1-\cos x} dx$ 2

(iv) Find $\int x^6 \ln x dx$ 2

(b) (i) Show that $\int_{-a}^a f(x) dx = \int_0^a [f(x) + f(-x)] dx$ 3

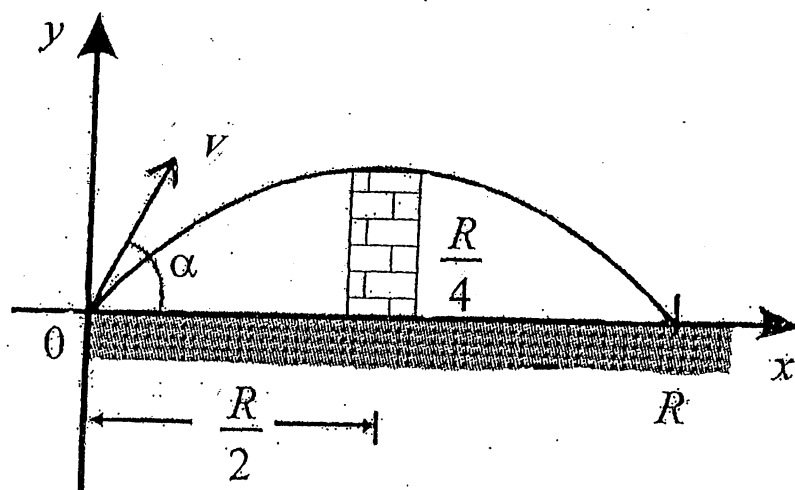
(ii) Hence show that $\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \frac{e^x \sin^2 3x}{1+e^x} dx = \frac{\pi}{6}$ 3

(c) (i) Let α and β be the roots of the equation $x^2 - ax + b = 0$.
Find the monic quadratic equation that has roots α^2 and β^2 . 2

(ii) Hence, show that the equation whose roots are the eighth powers of the roots of the equation $x^2 - x - 1 = 0$ is $x^2 - 47x + 1 = 0$. 3

(iii) Using the results of (ii) above show that $\left(\frac{47 + 21\sqrt{5}}{2} \right)^{\frac{1}{8}} = \frac{1 + \sqrt{5}}{2}$ 2

Question 8 (9 marks)



The diagram above shows a particle projected from a point O on horizontal ground with speed V m/s and at an angle of elevation, α° , where $0^\circ < \alpha^\circ < 90^\circ$.

- (i) Show that the *maximum* horizontal range of the particle is R where $R = \frac{V^2}{g}$. 4

You may assume that $\ddot{x} = 0$ and $\ddot{y} = -g$, where g is the acceleration due to gravity.
 x and y are the respective horizontal and vertical distances travelled by the particle.

- (ii) Show that the equation of the trajectory of the particle can be written as $y = x \tan \alpha^\circ - \frac{x^2 (1 + \tan^2 \alpha^\circ)}{2R}$. 1

You may **assume** that the formula for the trajectory is given by $y = x \tan \alpha^\circ - \frac{gx^2}{2V^2} (1 + \tan^2 \alpha^\circ)$.

- (iii) The particle just clears a wall, which has horizontal and vertical distances from O are $\frac{R}{2}$ and $\frac{R}{4}$ respectively.

(α) Find two possible value of α° . 2

(β) Find the horizontal range of the particle, in terms of R , for each possible value of α° . 2

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

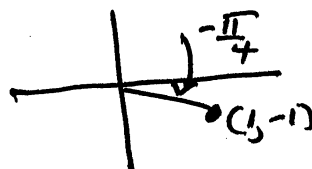
$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, x > 0$

QUESTION 1

$$\begin{aligned}
 (a) \quad z &= \frac{3-i}{2+i} = \frac{(3-i)(2-i)}{(2+i)(2-i)} \\
 &= \frac{6-5i-1}{5} \\
 &= \frac{5-5i}{5} \\
 &= 1-i
 \end{aligned}$$



$$\begin{aligned}
 (i) \quad \therefore z\bar{z} &= |z|^2 = (\sqrt{2})^2 \\
 &= \boxed{2} \quad \checkmark \checkmark
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad \tan(\arg z) &= \tan(-\frac{\pi}{4}) \\
 &= -\tan \frac{\pi}{4} \\
 &= \boxed{-1} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad (i) \quad z &= 1 + \cos \theta + i \sin \theta \\
 &= 1 + (2\cos^2 \frac{\theta}{2} - 1) + i \cdot 2\sin \frac{\theta}{2} \cdot \cos \frac{\theta}{2} \\
 &= 2\cos^2 \frac{\theta}{2} + 2i \sin \frac{\theta}{2} \cos \frac{\theta}{2} \\
 &= \underline{\underline{2\cos \frac{\theta}{2} (\cos \frac{\theta}{2} + i \sin \frac{\theta}{2})}} \quad \checkmark \checkmark \checkmark
 \end{aligned}$$

$$(ii) \quad \therefore |z| = \underline{\underline{2\cos \frac{\theta}{2}}} \quad \checkmark$$

$$\arg z = \underline{\underline{\frac{\theta}{2}}} \quad \checkmark$$

(in the form $r(\cos \phi + i \sin \phi)$
where $r = |z| = 2\cos \frac{\theta}{2}$
& $\phi = \frac{\theta}{2}$).

QUESTION 2.

(a)(i) Let $x = 2-x$ ie $x = 2-x$

$\therefore x^3 - 2x + 5 = 0$ becomes $(2-x)^3 - 2(2-x) + 5 = 0.$

$$\text{ie. } 8 - 12x + 6x^2 - x^3 - 4 + 2x + 5 = 0$$

$$\text{ie. } \boxed{x^3 - 6x^2 + 10x - 9 = 0}$$

(ii) Using the identity $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$
 $= 0^2 - 2 \times -2$
 $= 4.$

$$\alpha^2 + \beta^2, \beta^2 + \gamma^2 \text{ and } \gamma^2 + \alpha^2 \Rightarrow 4 - \gamma^2, 4 - \alpha^2, 4 - \beta^2$$

$$\text{Using } x = 4 - x^2 \Rightarrow x^2 = 4 - x \Rightarrow x = \sqrt{4-x}$$

$\therefore x^3 - 2x + 5 = 0$ becomes

$$(\sqrt{4-x})^3 - 2(\sqrt{4-x}) + 5 = 0$$

$$(4-x)\sqrt{4-x} - 2\sqrt{4-x} + 5 = 0$$

$$\sqrt{4-x}(4-x-2) = -5$$

$$\sqrt{4-x}(2-x) = -5$$

Squaring

$$(4-x)(4-4x+x^2) = 25$$

$$16 - 16x + 4x^2 - 4x + 4x^2 - x^3 = 25$$

Simplify becomes

$$\boxed{x^3 - 8x^2 + 20x + 9 = 0}$$

$$(b) (i) x^2 + (p+ri)x + q - ri = 0$$

$1 - 4i$ is a root.

$$\therefore (1-4i)^2 + (p+ri)(1-4i) + q - ri = 0.$$

$$-15 - 8i + p - 4pi + i + 4 + q - 5i = 0$$

$$-15 + p + 4 + q = 0$$

$$-8 - 4p + 1 - 5 = 0$$

$$p + q = 11$$

$$4p = -12$$

$$p = -3$$

$$q = 14.$$

$$(ii) \therefore x^2 + (-3+ri)x + 14-5i = 0$$

is the equation.

$$\sum \alpha = -(-3+ri) = 3-i$$

$$\therefore 1-4i + \gamma = 3-i$$

$$\boxed{\gamma = 2+3i}$$

(can't use
THE CONJUGATE
ROOT THEOREM.
as co-efficients
are not Real)

QUESTION 3.

$$(a) (i) w = \cos \theta + i \sin \theta.$$

$\therefore$ by De Moivre's theorem.

$$w^2 = \cos 2\theta + i \sin 2\theta.$$

$$\begin{aligned} \& \frac{1}{w} = w^{-1} = \cos(-\theta) + i \sin(-\theta) \\ & \text{OR } \cos \theta - i \sin \theta. \end{aligned} \quad \}$$

$$\begin{aligned} (ii) w^2 + 5w^{-1} - 2 &= \cos 2\theta + i \sin 2\theta + 5(\cos \theta - i \sin \theta) - 2. \\ &= 2\cos^2 \theta - 1 + 5\cos \theta - 2 + i(\sin 2\theta - 5\sin \theta) \end{aligned}$$

If this is purely imaginary the Real part is zero.

$$\text{i.e. } 2\cos^2 \theta + 5\cos \theta - 3 = 0. \quad (A)$$

$$\begin{aligned} (iii) \text{ Solving (A) } \cos \theta &= \frac{-5 \pm \sqrt{25 + 24}}{4} \quad \text{OR} \\ &= \underline{-3, \frac{1}{2}}. \quad (2\cos \theta - 1)(\cos \theta + 3) = 0 \\ & \qquad \qquad \qquad \cos \theta = \underline{\frac{1}{2}}, -3. \end{aligned}$$

$$\therefore \cos \theta = \frac{1}{2} \quad (\cos \theta \neq -3)$$

$$\boxed{\theta = \frac{\pi}{3}}$$

$$\therefore \underline{w = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}}$$

(B)

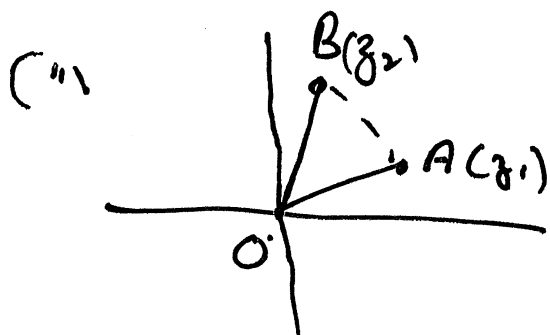
(b) (i) now $z_2 = \omega z_1$

$$\therefore \frac{z_2}{z_1} = \omega.$$

$$\text{now } \left| \frac{z_2}{z_1} \right| = |\omega| = \left| \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right|$$

$$\therefore \left| \frac{z_2}{z_1} \right| = 1. \quad \textcircled{C}$$

$$\left| \arg \frac{z_2}{z_1} = \arg \omega = \frac{\pi}{3} \right| \quad \left(\text{from a (iii), re. } \textcircled{B} \right).$$



now.

$$\left| \frac{z_2}{z_1} \right| = \frac{|z_2|}{|z_1|} = 1. \quad (\text{from } \textcircled{C})$$

$$\therefore |z_2| = |z_1|$$

$\therefore \triangle AOB$ is isosceles

Furthermore $\arg \frac{z_2}{z_1} = \arg z_2 - \arg z_1$

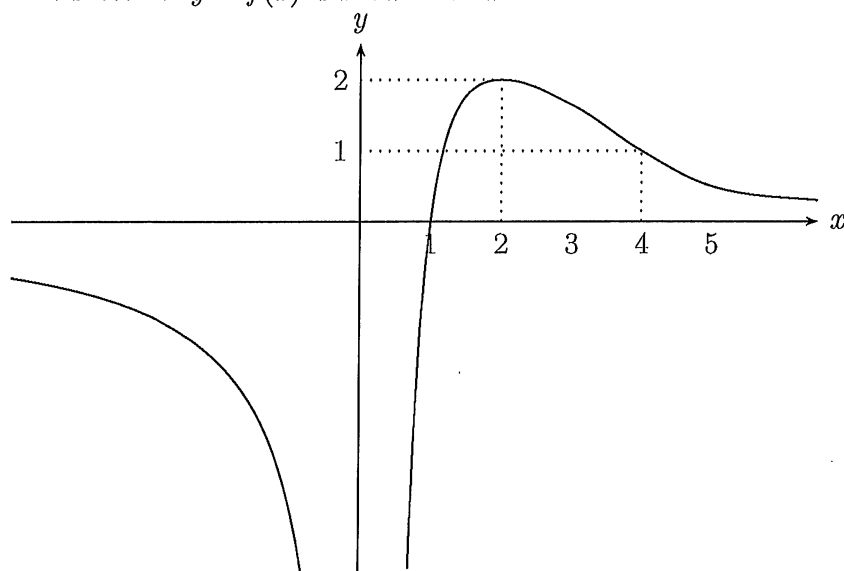
$$= \angle AOB$$

$$= \frac{\pi}{3}.$$

($\therefore \triangle AOB$ is equilateral as it is isosceles and contains a 60° angle!)

2006 Mathematics Extension 2 Assessment 2: Solutions Part B

4. (a) The sketch of $y = f(x)$ is shown below.

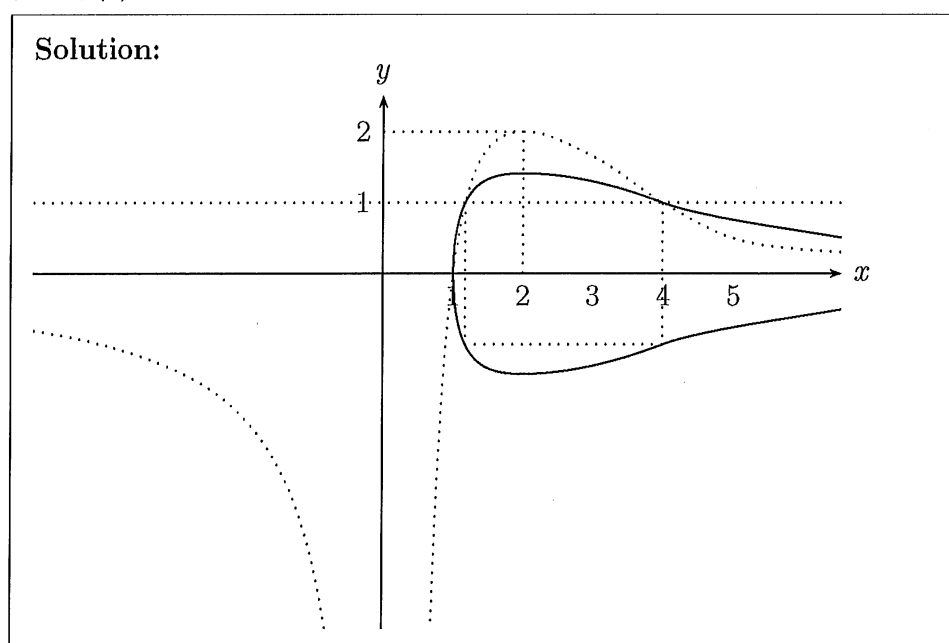


You are given the x -intercept is $x = 1$,
 $x = 0$ is a vertical asymptote and $y = 0$ is a horizontal asymptote.
 As well, $(2, 2)$ is a stationary point and $(4, 1)$ is a point of inflexion.

On the answer sheet provided draw separate graphs for

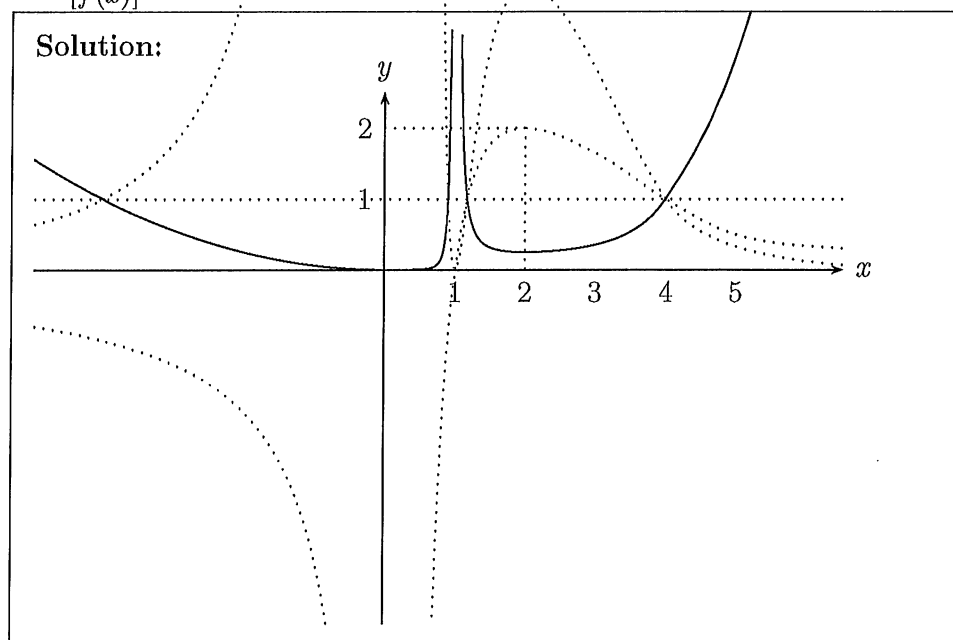
- (i) $y^2 = f(x)$

2



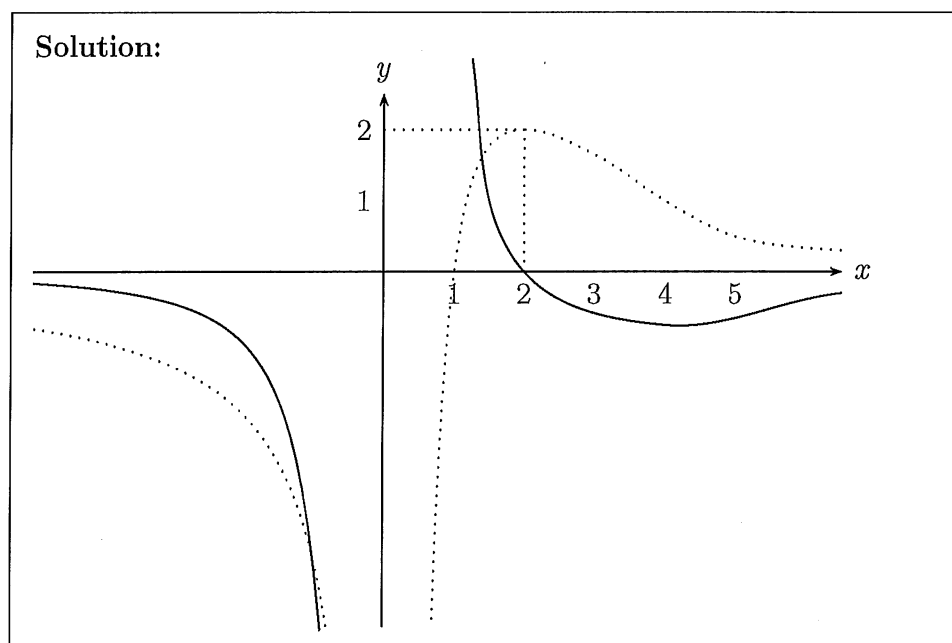
(ii) $y = \frac{1}{[f(x)]^2}$

3



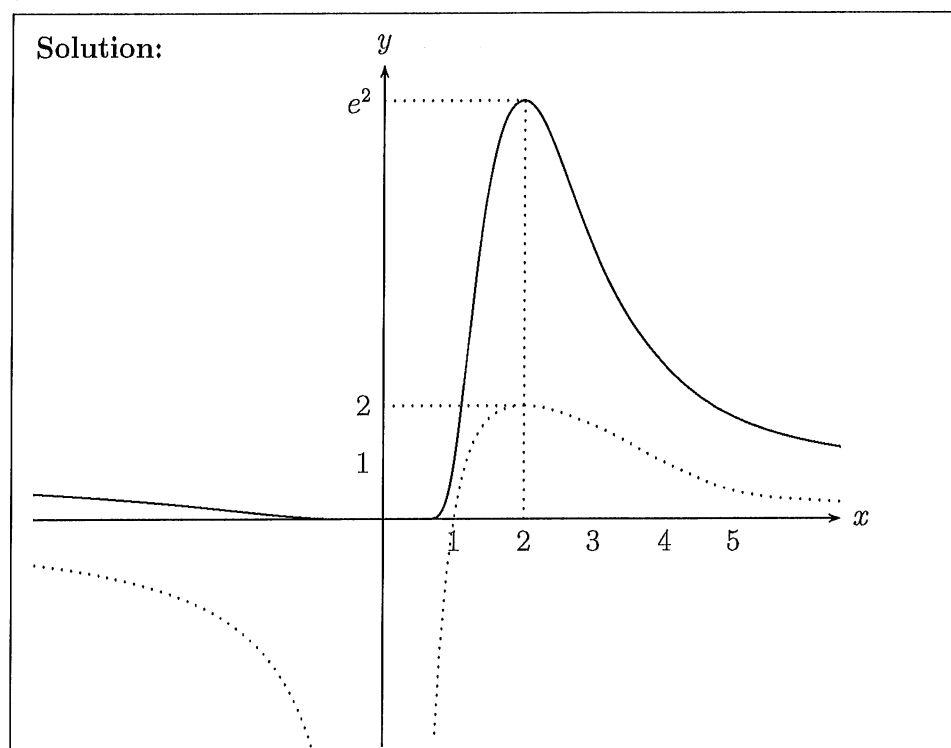
(iii) $y = f'(x)$

2



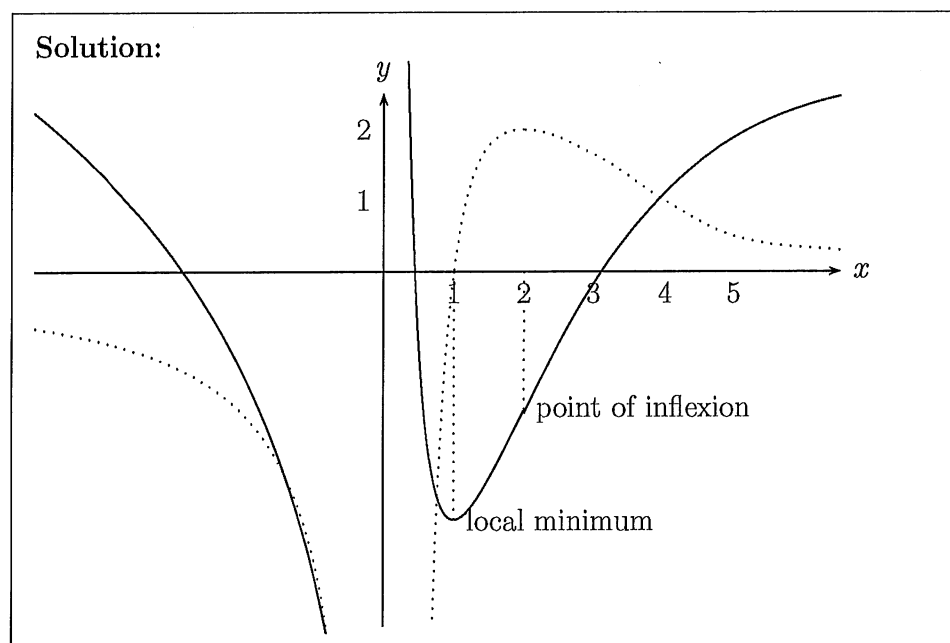
(iv) $y = e^{f(x)}$

2



(v) Draw a possible diagram for the primitive function of $f(x)$.

3



5. (a) Given that C is the curve defined by

$$f(x) = \frac{\sin 2x}{2 - \cos 2x}$$

where $0 \leq x \leq \pi$:

- (i) Find the x and y intercepts of C .

2

Solution: We note that $2 - \cos 2x > 0 \forall x$
 and $\sin 2x = 0$ when $2x = 0, \pi, 2\pi$,
 i.e., $x = 0, \frac{\pi}{2}, \pi$.
 So x -intercepts are at $0, \frac{\pi}{2}, \pi$
 and the y -intercept is 0.

- (ii) Find the turning point(s) of C and determine their nature.

3

Solution:

$$\begin{aligned} f'(x) &= \frac{(2 - \cos 2x)2 \cos 2x - \sin 2x(2 \sin 2x)}{(2 - \cos 2x)^2} \\ &= \frac{4 \cos 2x - 2 \cos^2 2x - 2 \sin^2 2x}{(2 - \cos 2x)^2} \\ &= \frac{2(2 \cos 2x - 1)}{(2 - \cos 2x)^2} \\ &= 0 \text{ when } \cos 2x = \frac{1}{2} \\ &\quad \text{i.e., } 2x = \frac{\pi}{3}, \frac{5\pi}{3} \\ &\quad \quad \quad x = \frac{\pi}{6}, \frac{5\pi}{6} \end{aligned}$$

Now

$$\begin{aligned} f'(0) &= \frac{2}{1} \\ f'\left(\frac{\pi}{2}\right) &= \frac{-6}{9} = -\frac{2}{3} \\ f'(\pi) &= \frac{2}{1} \end{aligned}$$

$\therefore$ Maximum turning point at $\left(\frac{\pi}{6}, \frac{1}{\sqrt{3}}\right)$
 and minimum turning point at $\left(\frac{5\pi}{6}, -\frac{1}{\sqrt{3}}\right)$.

- (iii) Show that $f(x) = -f(\pi - x)$.

2

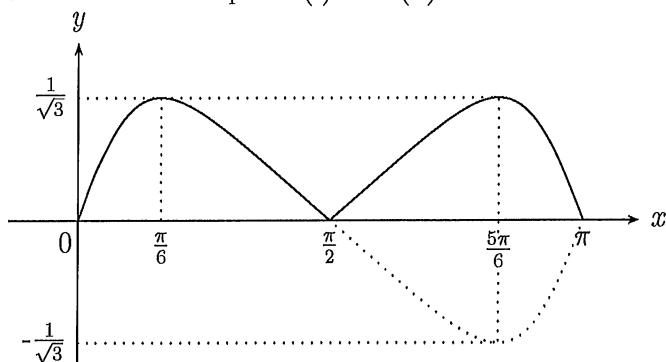
Solution:

$$\begin{aligned} \text{R.H.S.} &= \frac{-\sin(2[\pi - x])}{2 - \cos(2[\pi - x])} \\ &= \frac{-\sin(2\pi - 2x)}{2 - \cos(2\pi - 2x)} \\ &= \frac{-\sin(-2x)}{2 - \cos(-2x)} \\ &= \frac{\sin 2x}{2 - \cos 2x} \\ &= \text{L.H.S.} \end{aligned}$$

- (iv) Sketch the curve $y = \frac{|\sin 2x|}{2 - \cos 2x}$ for $0 \leq x \leq \pi$.

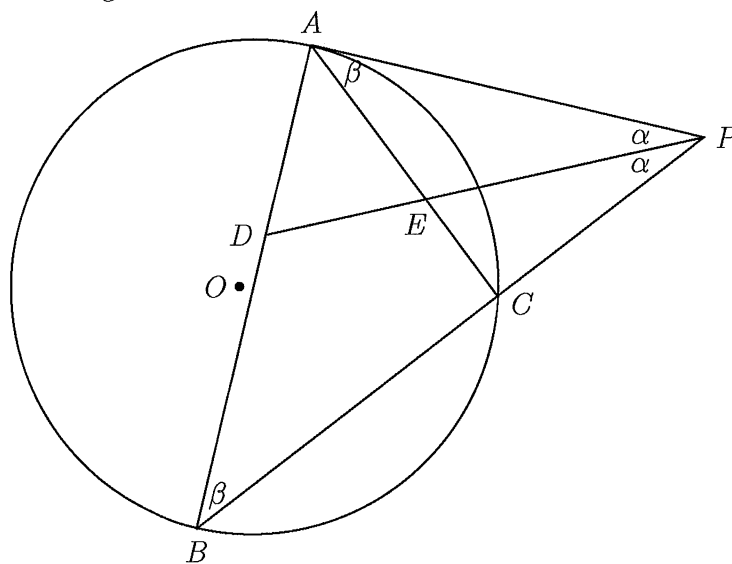
3

Solution: From parts (i) and (ii) we have



6. (a) In the diagram below, PA is the tangent to the circle at A , whose centre is O . Line PCB cuts the circle at B and C . The angle bisector of $\angle APB$ meets AB at D and AC at E .

6



Prove that $\frac{DB}{AB} + \frac{EC}{AC} = 1$.

Solution:

In $\triangle BAP$, ACP ,

$\angle APC = \angle BPA$ (common)

$\angle PAC = \angle PBA$ ($\angle$ in alternate segment)

$\therefore \triangle BAP \sim \triangle ACP$ (equiangular).

$\frac{BP}{AP} = \frac{BA}{AC}$ (corresp. sides of similar $\triangle$ s) [1]

[Solution continues on next page.]

Solution: [Continued from last page.]

Also, in $\triangle$ s DBP , EAP ,

$$\angle BPD = \angle APE \quad (PE \text{ bisects } \angle APB)$$

$$\angle PAC = \angle PBA \quad (\text{as above})$$

$\therefore \triangle DBP \sim \triangle EAP$ (equiangular).

$$\frac{BP}{AP} = \frac{DB}{EA} \quad (\text{corresp. sides of similar } \triangle\text{s}) \dots\dots\dots [2]$$

So, from [1] and [2] we have:

$$\frac{AC}{AB} = \frac{AE}{DB},$$

$$\frac{DB}{AB} = \frac{AE}{AC},$$

$$\frac{DB}{AB} = \frac{AC - EC}{AC},$$

$$\frac{DB}{AB} = 1 - \frac{EC}{AC},$$

$$\text{i.e., } \frac{DB}{AB} + \frac{EC}{AC} = 1.$$

- (b) How many ways are there to pick a man and a woman who are not husband and wife from a group of n married couples? [2]

Solution: Method 1—

Ways of choosing the man, $\binom{n}{1} = {}^nC_1 = n$.

Ways of choosing a woman who is not his wife = $n - 1$.

$\therefore$ Total number of ways = $n(n - 1)$.

Solution: Method 2—

Ways of choosing a couple without restriction = n^2 .

Ways of choosing a married couple = n .

$\therefore$ Total number of ways = $n^2 - n$.

Question 7

(a) (i) $\int \frac{e^x}{\sqrt{1-e^{2x}}} dx$ let $u = e^x$
 $du = e^x dx$

$$\therefore \int \frac{du}{\sqrt{1-u^2}} = \sin^{-1} u + C$$

$$= \sin^{-1}(e^x) + C$$

ii) $I = \int \frac{3x^2 - 6x + 1}{x^3 - 3x^2 + x - 3} dx = \log_e (x^3 - 3x^2 + x - 3) + C$

OR

$$3x^2 - 6x + 1 = A(x^2 + 1) + (Bx + C)(x - 3)$$

$$\Rightarrow A + B = 3 ; C - 3B = -6 ; A - 3C = 1$$

$$\therefore A = 1 \quad B = 2 \quad C = 0$$

$$\Rightarrow I = \int \left(\frac{1}{x+3} + \frac{2x}{x^2+1} \right) dx$$

$$= \ln(x+3) + \ln(x^2+1)$$

$$= \ln(x+3)(x^2+1)$$

$$= \ln(x^3 - 3x^2 + x - 3) + C$$

(iii) $\int \frac{1}{1 - \cos x} dx$ by "t results"

$$\Rightarrow t = \tan \frac{x}{2}$$

$$= \int \frac{\frac{2}{1+t^2} dt}{1 - \frac{1-t^2}{1+t^2}}$$

$$dt = \frac{1}{2} \sec^2 \frac{x}{2} dx$$

$$dt = \frac{1}{2} [1 + \tan^2 \frac{x}{2}] dx$$

$$= \int \frac{2}{1+t^2} dt$$

$$\frac{2dt}{1+t^2} = dx$$

$$= \int t^{-2} dt$$

$$= -\frac{1}{t}$$

$$= -\frac{1}{\tan \frac{x}{2}} \text{ or}$$

$$= -\cot \frac{x}{2} + C$$

(iv) $\int x^6 \ln x dx$

$$= \int \ln x \cdot \frac{d(x^7)}{dx} dx$$

$$= \frac{x^7}{7} \ln x - \int \frac{x^7}{7} \cdot \frac{1}{x} dx$$

$$= \frac{x^7}{7} \ln x - \frac{1}{7} \int x^6 dx$$

$$= \frac{x^7}{7} \ln x - \frac{1}{49} x^7 + C$$

(b)

Q(7) (b) LHS = $\int_{-a}^a f(x) dx$

(i)

$$= \int_{-a}^0 f(x) dx + \int_0^a f(x) dx$$

(let $x = -u$
etc.)

$$= \int_a^0 f(-u) \cdot -du + \int_0^a f(x) dx$$

$$= \int_0^a f(x) dx + \int_0^a f(x) dx$$

$$= \int_0^a [f(-x) + f(x)] dx$$

$$= \text{RHS}$$

(ii) $\int_{-\pi/3}^{\pi/3} \frac{e^x \sin^2 3x}{1+e^x} dx = \int_0^{\pi/3} \left(\frac{e^x \sin^2 3x}{1+e^x} + \frac{e^{-x} \sin^2(-3x)}{1+e^{-x}} \right) dx$

NOTE: $\sin^2(-3x) = [\sin(-3x)]^2$
 $= [-\sin 3x]^2$
 $= \sin^2 3x$

$$= \int_0^{\pi/3} \left(\frac{e^x \sin^2 3x}{1+e^x} + \frac{\sin^2 3x}{1+e^x} \right) dx$$

$$= \int_0^{\pi/3} \frac{\sin^2 3x \cdot (e^x + 1)}{(e^x + 1)} dx$$

$$\left(1 + e^{-x} = 1 + \frac{1}{e^x} \right)$$

$$= \int_0^{\pi/3} \sin^2 3x dx \quad \frac{1}{2}$$

$$= \int_0^{\pi/3} \frac{1}{2} (1 - \cos 6x) dx \quad \frac{1}{2}$$

$$= \frac{1}{2} \left[x - \frac{1}{6} \sin 6x \right]_0^{\pi/3} \quad \frac{1}{2}$$

$$= \frac{1}{2} \left[\left(\frac{\pi}{3} - 0 \right) - (0 - 0) \right] = \frac{\pi}{6} \quad \frac{1}{2}$$

$$(c) (i) \quad x^2 - ax + b = 0$$

$$\alpha + \beta = -\frac{(-a)}{1} = a \quad \frac{1}{2}$$

$$\alpha\beta = b$$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$\underline{\alpha^2 + \beta^2 = a^2 - 2b} \quad \frac{1}{2}$$

Required eqⁿ $x^2 - (\alpha^2 + \beta^2)x + \alpha^2\beta^2 = 0$

i.e. $x^2 - [a^2 - 2b]x + b^2 = 0 \quad |$

$$(ii) \quad x^2 - x - 1 = 0 \Rightarrow a=1 \quad b=-1 \quad \text{from (i)}$$

$$\therefore \text{Eq}^n \text{ with roots } \alpha^2, \beta^2 \Rightarrow x^2 - [1 - 2(-1)]x + (-1)^2 = 0$$

i.e. $x^2 - 3x + 1 = 0 \quad |$

$$\text{Eq}^n \text{ with roots } \alpha^4, \beta^4 \Rightarrow x^2 - [(-3)^2 - 2(1)]x + 1 = 0$$

i.e. $x^2 - 7x + 1 = 0 \quad |$

$$\text{Eq}^n \text{ with roots } \alpha^8, \beta^8 \Rightarrow x^2 - [(7)^2 - 2]x + 1 = 0$$

i.e. $x^2 - 47x + 1 = 0$

$$(ii) \quad x^2 - 47x + 1 = 0$$

$$x = \frac{47 \pm \sqrt{2205}}{2}$$

$$x = \frac{47 \pm \sqrt{21}}{2} \quad \frac{1}{2}$$

$$\therefore x = \frac{47 + \sqrt{21}}{2} \text{ applies}$$

$$\text{From } x^2 - x - 1 = 0$$

$$\Rightarrow x = \frac{1 \pm \sqrt{5}}{2} \quad \frac{1}{2}$$

$$\therefore x = \frac{1 + \sqrt{5}}{2} > 0 \text{ applies}$$

$$\text{SINCE } \frac{47 + 21\sqrt{5}}{2} = \left(\frac{1 + \sqrt{5}}{2} \right)^8 \Rightarrow \frac{1 + \sqrt{5}}{2} = \left(\frac{47 + 21\sqrt{5}}{2} \right)^{\frac{1}{8}}$$

Question 8

$t=0$

① $\ddot{x}=0 \quad \ddot{y}=-g \quad x=0 \quad y=0 \quad \dot{x}=V\cos\alpha \quad \dot{y}=V\sin\alpha$

$\dot{x}=C_1 \quad \dot{y}=-gt+C_2$

$\dot{x}=V\cos\alpha \quad C_2=V\sin\alpha$

$\dot{y}=V\sin\alpha-gt$

$x=Vt\cos\alpha + C_3$

$C_3=0$

$y=Vt\sin\alpha - \frac{gt^2}{2} + C_4$

①

$C_4=0$

$x=Vt\cos\alpha$

$y=Vt\sin\alpha - \frac{gt^2}{2}$

For max range let $y=0$

$\Rightarrow 0 = t(V\sin\alpha - \frac{gt}{2})$

$\therefore t=0 \text{ or } t = \frac{2V\sin\alpha}{g}$ ①

Range $= Vt\cos\alpha$ ✓
 $= V \cdot \frac{2V\sin\alpha}{g} \cdot \cos\alpha$

Range $= \frac{V^2}{g} \sin 2\alpha$ ✓ ①

R Max Range when $\alpha = \frac{\pi}{4}$

i.e. $R = \frac{V^2}{g}$ ①

(ii)

$y = x\tan\alpha - \frac{gx^2}{2V^2}(1+\tan^2\alpha)$

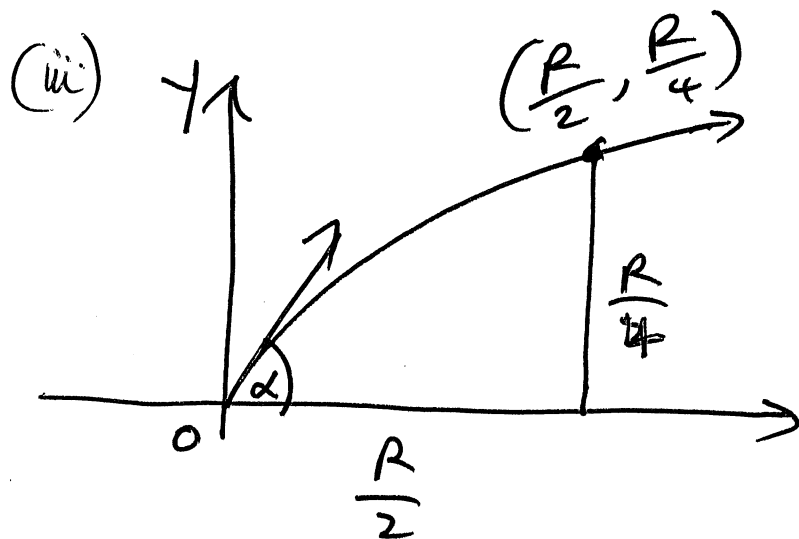
Since $R = \frac{V^2}{g}$

$\Rightarrow V^2 = Rg$ ①

$y = x\tan\alpha - \frac{gx^2}{2Rg}(1+\tan^2\alpha)$

i.e. $y = x\tan\alpha - \frac{x^2}{2R}(1+\tan^2\alpha)$

⊛



(x) Using (*) $\Rightarrow \frac{R}{4} = \frac{R}{2} \tan \alpha - \frac{\frac{R^2}{4}}{2R} (1 + \tan^2 \alpha)$

$$\frac{R}{4} = \frac{R}{2} \tan \alpha - \frac{R}{8} (1 + \tan^2 \alpha)$$

$$2 = 4 \tan \alpha - (1 + \tan^2 \alpha)$$

i.e. $\tan^2 \alpha - 4 \tan \alpha + 3 = 0$

$$(\tan \alpha - 3)(\tan \alpha + 1) = 0$$

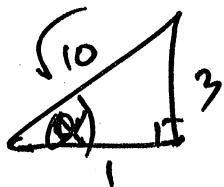
$$\tan \alpha = 3 \Rightarrow \boxed{\alpha \doteq 71^\circ 34'} \quad (1)$$

$$\tan \alpha = 1 \Rightarrow \boxed{\alpha = \frac{\pi}{4}} \quad (1)$$

(p) Range = $\frac{V^2}{g} \sin 2\alpha$

When $\theta = \frac{\pi}{4} \Rightarrow \boxed{\text{Range} = \frac{V^2}{g} \sin \frac{\pi}{2} = \frac{V^2}{g}} \quad (1)$

When $\theta = \tan^{-1} 3 \Rightarrow \text{Range} = \frac{V^2}{g} \cdot 2 \sin \alpha \cos \alpha \quad (1)$



$$\boxed{\text{Range} = \frac{V^2}{g} \cdot 2 \cdot \frac{3}{\sqrt{10}} \cdot \frac{1}{\sqrt{10}} = \frac{3}{5} R}$$